

Building Large-Scale U.S. Synthetic Electric Distribution System Models

Carlos Mateo^{ID}, Fernando Postigo, Fernando de Cuadra, Tomás Gómez San Roman, *Senior Member, IEEE*,
Tarek Elgindy, Pablo Dueñas, Bri-Mathias Hodge^{ID}, *Senior Member, IEEE*,
Venkat Krishnan^{ID}, *Senior Member, IEEE*, and Bryan Palmintier^{ID}, *Senior Member, IEEE*

Abstract—Rapid increases in distributed energy resources on distribution systems are prompting research efforts to improve and evaluate electric power distribution algorithms; however, there is a shortage of realistic, large-scale, U.S.-style test systems for the evaluation of such advanced algorithms. Some available tools to build large-scale test systems are of European style, and their application to studies focused on the United States might not be desirable given very different characteristics between the European and U.S. distribution designs. Motivated by this need, this paper develops detailed algorithms to build large-scale U.S. distribution systems and incorporates them in a new Reference Network Model, RNM-US. The approach starts with information from street maps and a catalog with electric equipment that includes power lines, transformers, voltage regulators, capacitors, and switching devices. The paper presents the algorithms through an illustrative case study of the different steps that comprise the process of building a synthetic distribution grid. Finally, the paper presents a medium- and a large-scale data set covering 10 million electrical nodes and 120,000 km of power lines, demonstrating the applicability of the proposed method to build very large-scale synthetic distribution systems.

Index Terms—Distribution system, synthetic system, test feeder, power utility, power flow, power system, reference network model, representative system.

I. INTRODUCTION

THE NEED for efficient integration of distributed energy resources (DERs) in distribution systems is prompting research efforts to improve and evaluate electric power distribution algorithms. A fair evaluation of these algorithms requires test systems that allow researchers to compare their

methods in realistic and diverse settings; however, publicly available information about real distribution systems is very limited because of concerns around privacy and revealing critical power infrastructure. These data are often available only to power utilities and require strict confidentiality agreements. Existing test feeders are limited to small- or medium-scale systems, which do not allow adequate testing of utility performance requirements. As a result, we identified the need to build large-scale synthetic distribution systems to cover this gap [1].

Previous test feeders typically cover only small portions of distribution systems, such as a single feeder. For example, the largest IEEE test system has only 8,500 nodes [2], whereas millions of nodes are found in real-world systems. Existing test systems are highly effective for their design purpose of confirming the accuracy and performance of distribution simulators, but they do not have the scale or detailed descriptions needed to test modern control algorithms. Building large-scale systems requires approximating the approaches taken by utilities in designing distribution systems and automating them such that they can be efficiently run for large regions.

As described in detail in Section II, however, past efforts at automation have primarily focused on one voltage level, and they do not provide a comprehensive method for planning entire distribution systems. Moreover, models usually choose from among several candidate solutions using mathematical programming, which can be difficult to scale to very large systems.

In previous papers, we presented the Reference Network Model for Europe (RNM-Europe) [3], [4] as well as assessments based on it [5]–[7]. U.S. distribution systems have unique features, however, such as the use of single-, two-, and three-phase laterals; the widespread deployment of voltage regulators and capacitors; and much smaller low voltage systems with center-tap transformers, among other structural differences [1]. These require a number of unique approaches and additional refinements to successfully build realistic U.S.-style systems.

Therefore, a major contribution of this paper is a detailed description of the new RNM algorithms tailored to automated planning of very large-scale U.S.-style distribution systems. Another important contribution is the step-by-step illustration of the methodology using a case study to show how to build the entire synthetic system. Such a detailed description of the algorithms and the step-by-step illustration of the methodology

Manuscript received January 8, 2020; revised May 2, 2020; accepted June 3, 2020. Date of publication June 19, 2020; date of current version October 21, 2020. This work was supported in part by the National Renewable Energy Laboratory, operated by Alliance for Sustainable Energy, LLC, for the U.S. Department of Energy (DOE) under Contract DE-AC36-08GO28308. Funding provided by Advanced Research Projects Agency-Energy (ARPA-E) under the GRID DATA Program. Paper no. TSG-00036-2020. (*Corresponding author: Carlos Mateo.*)

Carlos Mateo, Fernando Postigo, Fernando de Cuadra, and Tomás Gómez San Roman are with the Institute for Research in Technology (IIT), School of Engineering (ICAI), Universidad Pontificia Comillas, 28015 Madrid, Spain (e-mail: cmateo@comillas.edu).

Tarek Elgindy, Bri-Mathias Hodge, Venkat Krishnan, and Bryan Palmintier are with the Power Systems Engineering Center, National Renewable Energy Laboratory, Golden, CO 80401 USA.

Pablo Dueñas is with the Department of MIT Energy Initiative, Massachusetts Institute of Technology, Cambridge, MA 02139 USA.

Color versions of one or more of the figures in this article are available online at <http://ieeexplore.ieee.org>.

Digital Object Identifier 10.1109/TSG.2020.3001495

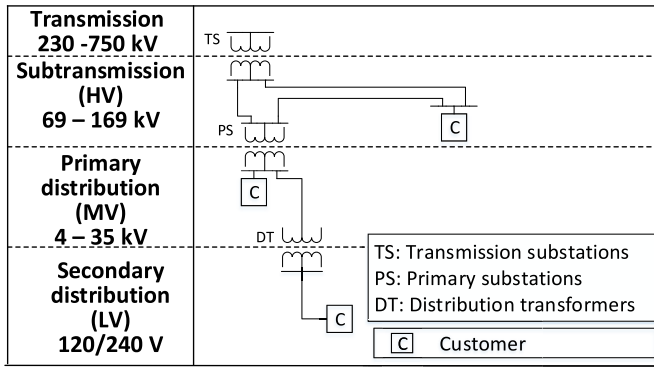


Fig. 1. Typical structure of distribution systems in the United States.

through a case study have not previously been presented for the RNM-Europe. We also showcase the potential of the algorithms to build realistic, very large-scale synthetic distribution systems by highlighting efforts to build distribution systems with millions of electrical nodes.

This paper is organized as follows. Section II provides a detailed review of relevant past efforts for automated distribution system planning. Section III highlights the differences between the RNM-US and the RNM-Europe. Section IV explains the methodology to build the synthetic distribution systems and validate them. Section V describes the algorithms applied in the RNM-US, which is the core of the proposed methodology. Section VI illustrates the algorithms using two case studies: a medium-size system for Greensboro, North Carolina (GSO), and a very large-scale system covering the San Francisco Bay Area, California (SFO). Finally, Section VII concludes the paper and proposes future lines of research.

II. BACKGROUND AND LITERATURE REVIEW

The typical structure of U.S. distribution systems is illustrated in Fig. 1. There are three major groups of voltage levels: high (HV), medium (MV), and low voltage (LV).

To build a synthetic distribution grid that covers a large region, planning algorithms for all voltage classes of the distribution grid are required. The objective of planning distribution systems is to locate and size substations and find the optimal way of connecting them to the customers while minimizing total installation and operation costs subject to technical (voltage and thermal limits) and geographic constraints (e.g., street maps) [8].

Distribution system planning algorithms can automatically design substations and feeders, locate control management devices, and install switching devices and tie-lines, as described in recent literature reviews [9]–[11]. There are two major trends in algorithms for planning distribution systems: mathematical programming and heuristics [1]. Mathematical programming usually guarantees optimality but typically requires more computation time, potentially considerably more. Heuristics are algorithms that can run much faster by aiming to find a good solution without ensuring obtaining

the global optimum. We apply heuristics in this paper to design very large-scale systems.

For substation planning, the decision variables are substation locations and sizes as well as their associated service areas [12]. Detailed analysis considering uncertainty can improve the robustness of the solution, but this is only viable when the size of the system is small [12]. Minimum spanning trees are the basis of substation location algorithms, typically followed by a stage of load clustering [13]. Distribution transformer planning has also been analyzed in the literature using Tabu Search [14] and k-means [15], among others. For substation and distribution transformer planning, we cluster the consumers by building a minimum spanning tree and removing branches of the graph because this allows us to separate the system into clusters that can be planned independently.

For feeder planning, the earliest technique was also the minimum spanning tree algorithm [16]. This algorithm finds the graph that connects all nodes while providing the minimum total branch length; however, this solution does not consider technical constraints (i.e., thermal and voltage limits). Common algorithms that are used to improve the initial solution are branch-exchange [17], evolutionary algorithms [16], and particle swarm optimization [18]. RNM-US uses the minimum spanning tree as the starting solution and then applies the branch-exchange because in this way we start with a low-cost solution and make it feasible. In addition, by modifying the branch-exchange algorithm, we can also include metric targets for the feeders. These are used to obtain results that are statistically similar to real U.S. distribution systems, as shown in [19].

For voltage management, voltage regulators and capacitors can be installed to improve voltages and energy losses [20]. For reliability, planning switches and tie-lines require combining the targets of cost minimization with reliability maximization [13]. This is usually a two-step process, wherein the structure of the distribution grid is first defined, and then the optimal number and location of the tie-lines are chosen. To assess reliability, the System Average Interruption Duration Index (SAIDI) and the System Average Interruption Frequency Index (SAIFI) can be calculated based on failure rates, switching device locations, and maintenance crew operations [21]. In RNM-US, we apply the approach described in [21] because it allows us to evaluate reliability for a synthetic distribution network planned by the model by using the parameters of the catalog of equipment and the layout of the network.

Schweitzer *et al.* propose a method to generate synthetic distribution systems by aiming to replicate graph parameters such as the node degree or the distance to the source and by considering electrical constraints such as voltage drop and the thermal limits of the cables [22]; however, the authors focus exclusively on the medium voltage system, the solutions are purely radial (without loops or meshes), and the graphic representations of the obtained systems are graphs that lack a topological description of the system that considers street maps. The authors apply this methodology to a European distribution system in the Netherlands. Moreira *et al.* propose an optimization algorithm based on branch-exchange to optimize the layout of the system and use dynamic programming to

size conductors [23]; however, their case example has only 1,088 consumers, which is several magnitudes lower than typical utility systems. Also, their systems cover only low voltage cases, are purely radial, and do not follow the street map. Pisano *et al.* propose a method to generate synthetic distribution system models using open data and geo-referenced information [24]. They have stages to define demand and generation, and they associate them with the substations. Their design of power lines is highly simplified, however, using only a combination of four typical feeders. They apply the methodology to an Italian case study. Gan *et al.* propose using fractals to generate synthetic distribution systems [25]; however, they focus exclusively on the low voltage level, modeling U.K. distribution systems, and they do not consider the street map. Ignoring street maps not only decreases the realism of the layouts but also has implications on the total length—and cost—of the systems [3]. Saha *et al.* cite the model that we propose in this paper and describe another framework to build synthetic distribution grids [26]. They use street maps, but their obtained system seems to be biased toward horizontal and vertical lines. They focus only on the substations and the medium voltage system, considering multiphase systems, but the paper lacks the coherence stage of our model, which prevents having three-phase power lines downstream of single-phase laterals [27].

Recently, Grzanic *et al.* have developed the Distribution Network Models platform (DiNeMo) to build European representative systems using the previously developed RNM-Europe as the core [28]. Alternative existing Reference Network Models are limited to the Network Performance Assessment Model (NPAM) [29] and a System for the Automatic Generation of Theoretical Network Models (ANETO) [30], both also developed in the context of regulation. NPAM, however, does not build synthetic distribution systems directly; instead, it simply estimates the corresponding costs. ANETO builds a synthetic grid and covers several voltage levels, but it is not adapted to the United States.

As shown in Section VI, the RNM-US model described in this paper has been applied to build very large-scale distribution systems with nearly 10 million electric nodes, and it has been validated exhaustively from several points of view [19]. These distribution systems have been publicly released to be available to other researchers to test and benchmark their algorithms. These distribution systems show the unique features of the proposed model, which is capable of building very large-scale systems.

III. RNM-US vs. RNM-EUROPE

Broadly speaking, worldwide there are two main types of designs for distribution systems: European and U.S., [31]. This is because of historic, socioeconomic, and cultural reasons. Neither is *a priori* better than the other; however, the many differences have required developing a U.S.-specific RNM.

RNM-US is coded in C++ and uses the same layer architecture as RNM-Europe [3], including: (1) a logical layer that defines the basic system structure—nodes, branches, graphs,

etc.—and implements, among others, the minimum spanning tree using Prim's algorithm; (2) a topological layer with geospatial information and algorithms within the Geographic Information System (GIS) to find the optimal layout of power lines while considering street maps; (3) an electrical layer comprising a set of electrical equipment with attributes as well as power flow and planning algorithms; and (4) a quality of supply layer to evaluate the reliability indices, including algorithms to improve them. This basic architecture is generic and can be used both in the European and the U.S. contexts; however, the planning algorithms needed to be rebuilt entirely, reusing only basic concepts from the previous model. The major differences are highlighted next.

A. Overhead/Underground Modeling and Street Maps

Distribution systems use typically a combination of underground and overhead installations. In RNM-Europe, the model considers that rural areas are overhead, whereas underground installations are typically used in the center of towns. In RNM-US, the underground/overhead attribute of each consumer is instead provided as an input to the model based on the average height of the buildings in an area. Typically, underground cables in the United States are used in downtown urban areas where high-rise buildings are located and in newer suburban developments. Moreover, in RNM-Europe we used a synthetic street map, automatically generated by the model [3], whereas in RNM-US we use real street maps.

B. Low Voltage System (Secondaries) and Distribution Transformers

The low voltage system is different in Europe and the United States in several ways. First, the low voltage system in the United States supplies very few consumers from each distribution transformer (usually 10 or less), whereas in Europe distribution transformers connect hundreds of consumers. Therefore, the low voltage system downstream of each distribution transformer has a different structure and is much shorter in the United States. European low voltage systems also use higher voltages and might include extensive building mounting.

In Europe, three-phase distribution transformers are commonly used, connecting not only three-phase consumers but also single-phase consumers that need to be distributed along the phases to balance the system. In the United States, typically three-phase distribution transformers connect only three-phase consumers, whereas single-phase consumers are typically connected to single-phase center-tap transformers; hence, not only the nominal voltages of the transformers are different, but also the electrical connection is different. The specificities of the single-phase center-tap transformers have been modeled in RNM-US, providing the electrical connectivity in the OpenDSS format.

C. Medium Voltage System

In Europe, distribution systems are three-phase in all voltage levels, whereas in the United States single-, two-, and three-phase primary feeders coexist. Therefore, not only is

consumption unbalanced but also the system structure itself is unbalanced, with single- and two-phase laterals extending primaries to points closer to the final consumers. This has required specific algorithms for designing the phases of the feeders [27]. This has also made it necessary to change the output of the RNM that is used to run power flows. In RNM-Europe, we provide the output in MATPOWER format, which is simpler but only allows running three-phase balanced power flows. Instead, in RNM-US the output for running power flows is provided in OpenDSS format to enable specifying the phases of feeders and consumers.

Voltage regulators and capacitor banks were rarely used in RNM-Europe. In RNM-Europe, voltage regulators were used only as a remedial action in certain rural areas with serious voltage problems. U.S. distribution systems, however, use alternatively (and sometimes simultaneously) capacitor banks and voltage regulators to control voltage, with high penetrations of these devices scattered throughout primaries. Therefore, RNM-US has been designed to consider them as a standard planning option, not only as a casual solution. Moreover, statistics of the penetration of these devices in U.S. distribution systems have been considered to plan these devices in RNM-US [19].

D. Sub-Transmission System and Transmission Substations

In RNM-Europe, transmission substations were an input to the model, so their location was known and fixed. In RNM-US, this information was typically not available, and therefore we have developed another planning stage for locating and sizing them using basic concepts of RNM-Europe. We have modified the design of the sub-transmission system to make it compatible with the proposed transmission substations. In case feasibility is not achieved, there is an iterative process between the sub-transmission system planning and the algorithm that locates transmission substations.

E. Validation and Metrics

In RNM-Europe, results were manually contrasted against data from the Distribution System Operators. A validation process was performed in the DSO Observatory for three specific systems [32] using the 0.05 percentile, the median, and the 0.95 percentile of nine metrics. These systems were obtained with a manual iterative process applied specifically to these three representative systems. In RNM-US, a more exhaustive validation has been applied. It uses a three-pronged validation process that includes statistical, operational, and expert analysis and feedback. In particular, the operational validation checks parameters including voltage drop, energy losses, and imbalance levels. The expert feedback has been extremely useful to understand the architecture of real U.S. distribution systems, and the statistical validation has verified the histograms of the synthetic distribution systems built with RNM-US by comparing them with the corresponding histograms of more than 10,000 real feeders from U.S. utilities [19]. To be able to obtain a positive validation, specific metrics have been introduced in the model as targets to



Fig. 2. Differences in urban areas between (a) RNM-Europe (left) and (b) RNM-US (right).

ensure that the obtained synthetic distribution systems have similar characteristics to real U.S. distribution systems [19].

The electrical planning algorithms of RNM-US have been rebuilt from scratch to model these differential characteristics of the U.S. distribution system. Fig. 2 illustrates the graphic differences in the systems built using RNM-Europe [3] (on the left) and RNM-US (on the right). One clear difference is the number of phases. In RNM-Europe (on the left), all the systems are three-phase (represented by black). On the right, in RNM-US, three-phase lines (represented by black) are combined with single-phase lines corresponding to phases A, B, C and represented by red, green, and blue, respectively. In addition, the secondaries (thin lines) are significantly different. In RNM-US, the buildings (connection points) are located inside of city blocks, and there is a low voltage power line connecting each distribution transformer in the street to the respective building that it supplies. In addition to these graphic and structural differences, the statistics of the systems are significantly different.

IV. SYNTHETIC DISTRIBUTION SYSTEM CREATION METHODOLOGY

The methodology applied to build synthetic distribution systems is illustrated in Fig. 3. The core of the methodology in the RNM-US is able to design distribution grids to supply electricity to consumers based on electrical, geographic, and reliability constraints. The main inputs of the model are:

- Consumers (location, demand, and number of phases)
- Street maps (geographic constraint)
- A catalog of electric equipment: power lines, transformers, capacitors, etc., with their technical and economic parameters
- A set of configuration parameters (design criteria, voltage thresholds, metric targets). Some examples of metric targets are the length of the feeders, the total installed capacity of the distribution transformers per feeder, and the number of consumers per feeder.

The process starts with parcel or land use information (typically from commercial vendors) and building footprint and height data (often publicly available in OpenStreetMaps). This information is used for labeling with a category (e.g., single- or multifamily household, supermarket, hotel, restaurant, or school) and estimating the location of the consumers. Because each building lies within a parcel, the category is obtained after intersecting the building centroid with each parcel polygon.

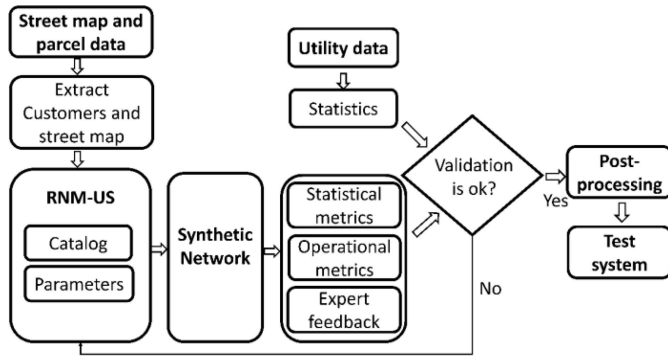


Fig. 3. Simplified block diagram of RNM-US methodology.

Assuming that the energy consumption is correlated with the building volume, the peak load is obtained through linear interpolation by using a set of reference building models [33], [34]. The system layout follows the layout of streets, which are obtained from publicly available street maps. The information is complemented with a catalog of components and design criteria based on the analysis of actual data about the utilities. The main outputs are:

- Geographic Information System (GIS) files in ESRI Shapefile format to enable the graphic display of the system
- A description of the system in OpenDSS and CYME to enable running power flows [35], [36]
- CSV files describing the system components
- A summary file, with the main system metrics and feeder statistics.

The RNM-US output is fed into the National Renewable Energy Laboratory's DiTTo¹ converter to transform the OpenDSS description of the system into CYME and post-process it. Post-processing is used to enhance some details of the base electrical system from RNM-US, including adding specific control schemes for voltage regulators and capacitors, detailing the multi-transformer bank arrangement inside substations, adjusting loads and voltage regulating schemes as needed, and adding time-series loads and rich DER scenarios.

As shown in Fig. 3, RNM-US is calibrated by validating the synthetic data sets it produces. The validation approach uses real-world utility data and includes a three-pronged process [19], as summarized in Fig. 4, comprising:

- *Statistical validation:* Histograms of characteristic metrics in the test systems are compared against the same metrics' histograms found in the real utility data. The metric histograms are compared to ensure that synthetic systems capture the appropriate level of variety found in real-world metrics.
- *Operational validation:* Power flows and voltage profiles are analyzed using OpenDSS and vendor tools CYME and EDD.
- *Expert validation:* Updates are made to system design based on the feedback provided by industry experts.

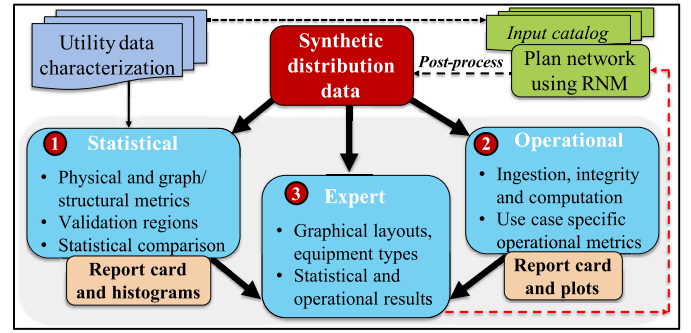


Fig. 4. Three-pronged validation process: an overview.

This validation process typically results in modifications to the RNM-US algorithm or input catalog data. The process is iterated several times until the characteristic metrics from RNM-US match the characteristic metrics from utility data (checked during the validation process), the power flows look realistic, and the modifications suggested by the experts are implemented. The outcome of this process is eventually a validated large-scale distribution test system from the RNM-US. This paper focuses on the algorithms that have been implemented in the RNM-US for building synthetic distribution systems. Details on the validation process, list of metrics from synthetic systems that were compared against real utility data, and a detailed explanation of validation results are documented in [19].

V. ALGORITHMS IN RNM-US

A. Flowchart

The flowchart in Fig. 5 describes the steps used to build the entire synthetic distribution system using RNM-US. The algorithms of the RNM-US correspond roughly to the structure of the distribution grids themselves.

The model starts bottom-up from the consumers and the lowest voltage level, and it proceeds upstream to the sub-transmission level. The planning of the active and reactive power (kW and kVAr) of the loads at each voltage level are provided as inputs to the algorithm. At each voltage level, the model starts planning the electricity supply sources (i.e., substations or distribution transformers). As described in the following sections, the model then plans the lines required to connect the sources to the loads. Every time the system planning algorithm proposes connecting two nodes, it interacts with the GIS to determine the optimal layout of power lines that complies with the street map. For computational tractability, the system construction process is partitioned into separate processes for each voltage class.

B. Low Voltage Level

1) *Distribution Transformers:* In LV, the sources are the distribution transformers, and the loads are LV consumers. The model first clusters the consumers by building a minimum spanning tree and removing branches of the graph (see Fig. 6) [13]. The algorithm removes the branches while checking that the resulting clusters are feasible. This process aims to

¹<https://github.com/NREL/ditto>

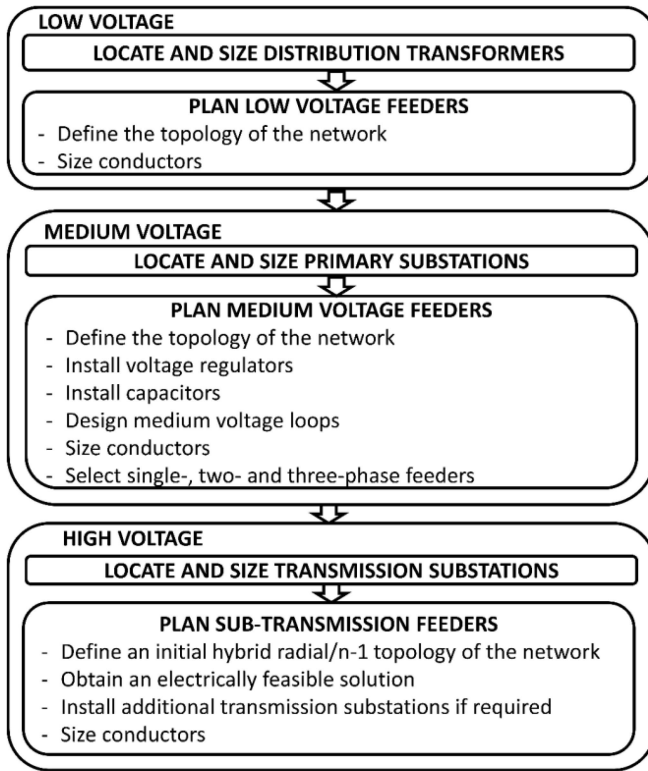


Fig. 5. RNM-US algorithms flowchart.



Fig. 6. Consumer clustering to estimate distribution transformer and substation location.

remove long branches to make smaller clusters (each fed from a different distribution transformer), reduce cost, and be able to analyze and build each cluster separately. The feasibility constraints that are verified are the voltage and thermal limits. A distribution transformer is set on the closest street map point to each centroid because the distribution transformer is the source for all the loads within the cluster. The size of the distribution transformer is calculated based on the simultaneous peak demand of the consumers [3] and on using an oversizing margin.² For single-phase consumers, the model installs center-tap transformers typically used in U.S. systems.

2) *Low Voltage System*: Once both the sources and the loads are known, the model starts laying out the LV lines. In the United States, LV systems usually connect only 10 or fewer consumers [37], and they are commonly designed to have a tree or star configuration. The structure of the secondaries that connect the distribution transformers to the LV consumers is illustrated in Fig. 7. The algorithm to obtain the secondaries is based on the minimum spanning tree, with the nodes being not only the consumers and the distribution

²The transformers are sized to withstand double the simultaneous peak demand, according to the practice observed in U.S. utilities. A simultaneity factor lower than 1.0 is also applied.

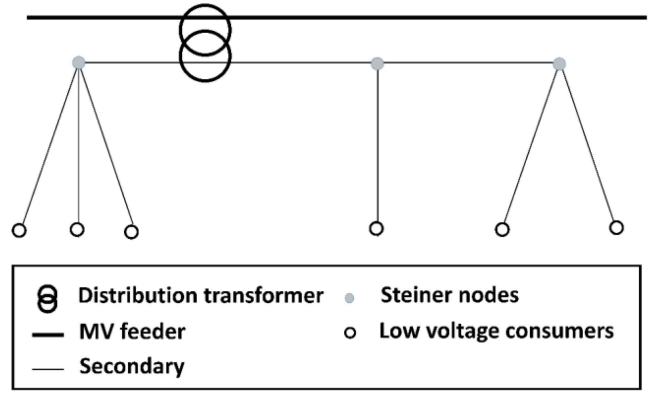


Fig. 7. Low voltage configuration.

transformer but also additional Steiner nodes [38], located in the streets. This configuration allows representing more situations than the two basic configurations, the pure star (from the distribution transformer directly to the buildings) and the pure tree (without stars). These two basic configurations are special cases of the proposed one, and the proposed solution represents the additional complexity observed in real systems.

C. Medium Voltage Level

1) *Primary Substations*: The process is applied in a similar way for the MV portion of the system. Primary substations are placed first, then the feeders are constructed. When designing clusters and planning the primary substations, the estimated voltage drop and installed capacity are evaluated against the corresponding constraints (voltage and thermal limits) as a feasibility criterion. Additionally, metrics such as the expected number of consumers per feeder are also considered. The resulting substations are constrained by the target values for these metrics to prevent having too long feeders, too much capacity of the distribution transformers per feeder, or too many consumers per feeder. These are used to obtain results that are statistically similar to real U.S. distribution systems, as shown in [19].

2) *Medium Voltage Feeders*: In MV, the topology of the feeder starts with the minimum spanning tree solution [16], and then a branch-exchange algorithm is applied [17]. The voltage limits considered in this stage are relaxed because voltage regulators and capacitors are later installed to improve voltages and comply with the ANSI limits.

In the United States, MV feeders can have single-, two-, or three-phase segments. This is considered by the algorithms, with a stage that decides the number of phases according to technical and economic criteria, as described in [27]. Then, it chooses phases A, B, and C, with the objective function of minimizing voltage unbalances, by swapping the downstream phases of every node and selecting the ones that result in a lower unbalance.

3) *Voltage Control Devices*: Voltage regulators and capacitors also need to be allocated on the MV system. Fig. 8 illustrates how voltage regulators are placed to avoid voltages exceeding limits. The voltage typically starts dropping on the feeder as the distance from the substation increases. When

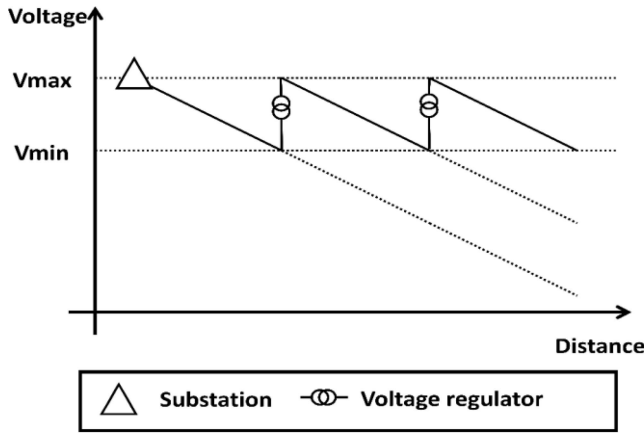


Fig. 8. Voltage regulators.

voltage is low, a voltage regulator is installed to increase it, to avoid an infeasible solution. This strategy is implemented with a recursive algorithm using a Depth-First Search to analyze the system. Starting with the nodes closer to the substation, voltage regulators are installed in the first node that exceeds certain thresholds (i.e., half the maximum allowed voltage range) if there are several problems downstream. This aims to solve several downstream problems with a voltage regulator located upstream. If that is not the case, a voltage regulator is placed in the vicinity of the nodes with voltage violations to solve them.

Likewise, for capacitors, the objective is also to improve voltages; however, the function that guides where capacitors are installed seeks to minimize energy losses. Given that the losses are related to voltage drop over the line, the objectives to minimize losses and improve voltages are closely interrelated. Losses in a single-phase power line when the capacitor is located at the end of the line can be estimated according to Eq. (1).

$$L = \frac{(\Delta V)^2}{Z^2} R \quad (1)$$

where L are the power losses, ΔV is the voltage drop, Z is the impedance of the feeder, and R is the resistance of the feeder.

For a fixed system structure and assuming only one type of conductor, the factor that relates resistance and impedance is constant, and the minimization of losses leads to a minimization of voltage drop and vice versa. Therefore, this idea is applied to target the minimization of energy losses while seeking an improvement in voltages. This assumption is in line with other research papers that allocate capacitors to minimize energy losses [39]. The stopping criterion is based on a ratio that estimates the percentage of the cost of the capacitor that is recovered by the reduction of energy losses while also considering the target metrics.

4) *Loops and Switching Devices*: The MV system also requires designing tie-lines to facilitate reconfiguration. There are three possible types of loops [40]:

- Loops between different substations
- Loops between different feeders of the same substation
- Loops within the same feeder.

The objective function that selects the best candidates to install tie-lines maximizes the length of the meshed section

and minimizes the length of the new tie-line. This aims to maximize reliability (increasing the size of the meshed section) with the minimum possible cost (minimum length of the new branch). In the case of loops within the same feeder, the maximum length of the loop is constrained. Breakers are installed in the substations. Switches and reclosers are also installed in the MV system to improve reliability by identifying the most critical branches in terms of upstream demand and downstream failure rate. The stop criteria is based on target metrics and on SAIDI and SAIFI constraints, which are estimated by simulating how the maintenance crews locate and isolate faults and restore service [21].

D. High Voltage Level

1) *Transmission Substations*: The transmission substations are the interface between transmission and distribution. Transmission substations can be either known a priori or decided by the model. When the transmission substations are decided by the model, RNM-US applies the algorithm described in Fig. 6 to place and size them. In this case, the primary substations and the HV consumers are the loads of this voltage level. Alternatively, it is possible to provide the actual transmission substations as inputs.

2) *Sub-Transmission Networks*: In the case of the sub-transmission system, the objectives are cost minimization and reliability improvement. The algorithm first starts with a radial configuration that is used as an initial low-cost solution. This initial radial configuration is based on the minimum spanning tree. Then, meshed branches are added to target an “n-1” configuration and improve reliability while allowing some sections to remain radial (when the reliability improvement is not significant). Finally, electrical constraints are solved by adding more power lines to achieve feasibility in terms of thermal and voltage limits. For this purpose, the algorithm proposes candidate power lines based on a Delaunay triangulation [41]. The candidates are first ranked using a heuristic function ($f1$). The use of the heuristic to rank candidates significantly reduces the number of solutions to check and thus the total computing time. Only the best candidates are tested and compared to each other by running a power flow for each of them and using an objective function ($f2$) to evaluate them. The best one (according to function $f2$) is added to the system. The algorithm works iteratively and stops adding candidates when the system is electrically feasible. If the algorithm does not converge, more transmission substations are installed, until an electrically feasible solution is found.

The objective (minimization) function ($f1$) that guides the heuristic to rank the candidates is shown in Eq. (2). This function is defined to minimize the length of the extra branch and to maximize the existing violation solved (e.g., excess of voltage over the ANSI limit or power flow over the thermal limit). The electrical distance between the two nodes is also in the denominator to connect nodes that are electrically far away.

$$\text{Min } f1 = \frac{\text{len}_{b_{n1,n2}}}{VVED_{n1,n2} + VVED_{n2,n1} + CVED_{n1,n2} + CVED_{n2,n1}} \quad (2)$$

where $f1$ is the objective function of the heuristic that ranks the solutions; $VVED_{n1,n2}$ is the voltage violation of node $n1$ multiplied by the electrical distance between nodes $n1$ and $n2$ (measured with voltage magnitudes); and $CVED_{n1,n2}$ is the thermal limit violation of the branches connected to node $n1$ multiplied by the electrical distance between nodes $n1$ and $n2$ (measured with voltage angles).

The objective (minimization) function ($f2$) that finally selects the best solution is defined by Eq. (3) to minimize the length of the new branch and the resulting voltage and thermal violation. In this function, the violations are in the numerator because Eq. (3) checks the violations after adding the branch, whereas Eq. (2) checks the violations before adding the branch.

$$\text{Min } f2 = \text{len}_{b_{n1,n2}} \times \left(\sum_{n_i} OV_{n_i} + \sum_{b_{n_i,n_j}} CO_{b_{n1,n2}} \right) \quad (3)$$

where $f2$ is the objective function that determines whether the solution is effective to solve the violations at minimum cost; $OV_{n1}(p.u.)$ is the overvoltage or undervoltage of node $n1$ in per unit; and $CO_{b_{n1,n2}}(p.u.)$ is the thermal limit overload of branch $b_{n1,n2}$.

VI. CASE STUDIES

A. Medium-Scale System: Greensboro, North Carolina

The proposed methodology is applied to generate a synthetic distribution system in Greensboro, North Carolina. The system covers three types of areas: urban-suburban, industrial, and rural, which are generated separately. The following illustrations of the RNM-US algorithms show a zoom-in to the urban-suburban area, which corresponds to a densely populated residential area.

As discussed in Section IV, the main inputs of the RNM-US are the location and demand of the consumers, a street map, a catalog of equipment, and a set of configuration parameters. Fig. 9 shows the location of the consumers as well as the street map. The closest connection points of the buildings to the streets represent the consumers, whereas the street map is built starting with a series of points that represent potential service drops.

The model starts from the lowest voltage level. The first step is sizing and locating distribution transformers, as shown in Fig. 10. For this purpose, a minimum spanning tree is computed to connect the consumers [16]. Branches of the minimum spanning tree are removed to separate the consumers into clusters [13]. This stage verifies that supplying each cluster with a single distribution transformer is feasible. The transformer is placed in the closest point of the street map to the cluster centroid. The distribution transformers are selected to supply the demand of the consumers of the cluster, with a given oversizing margin.

In the LV system, distribution transformers are the sources, and LV consumers are the loads. In this step, shown in Fig. 11, each cluster of consumers is connected to its distribution transformer via the LV lines (i.e., the secondaries). This system has a tree configuration, as shown in Fig. 7, with

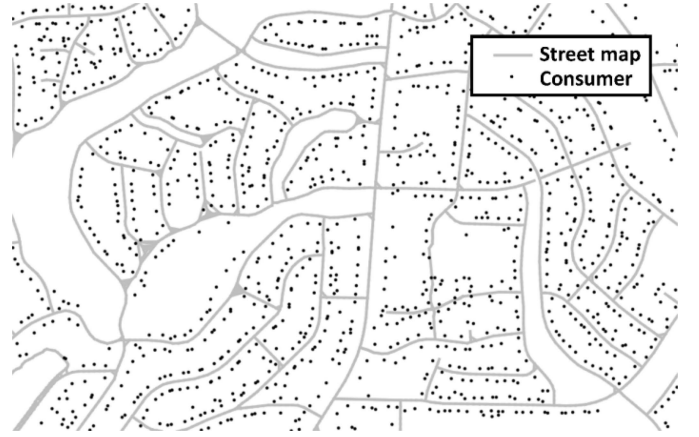


Fig. 9. Consumers and street map.

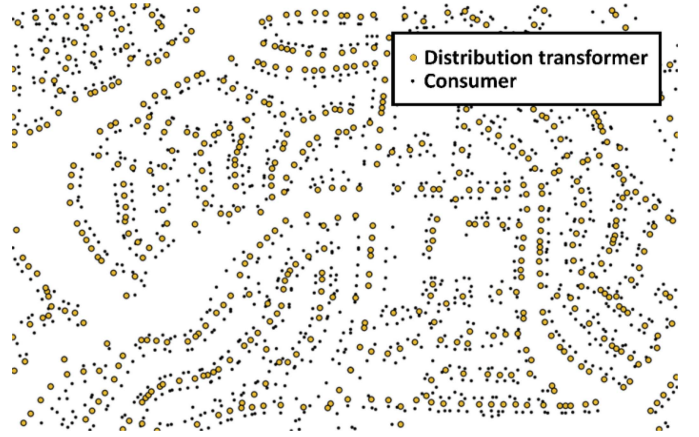


Fig. 10. Distribution transformers.

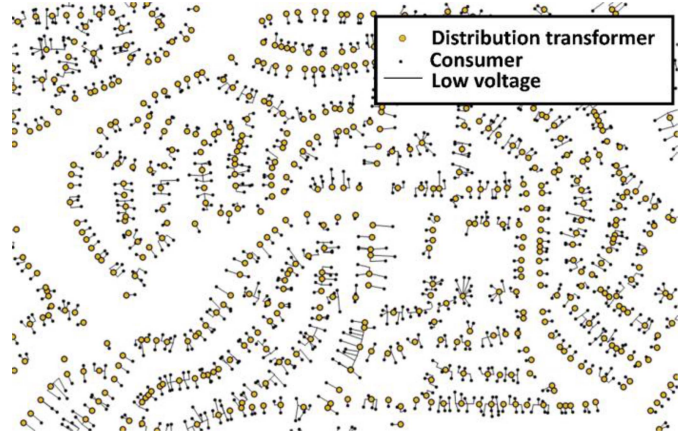


Fig. 11. Low voltage system.

intermediate (Steiner) nodes [38]. In certain cases, two consumers can be very close to each other, resulting in a star connection.

Fig. 12 shows the planning of the primary substations. In this case, the loads of the MV system are the distribution transformers and MV consumers. The process is analogous to the distribution transformer planning phase: (a) obtain a minimum spanning tree connecting the loads, and (b) remove branches to obtain clusters [13].

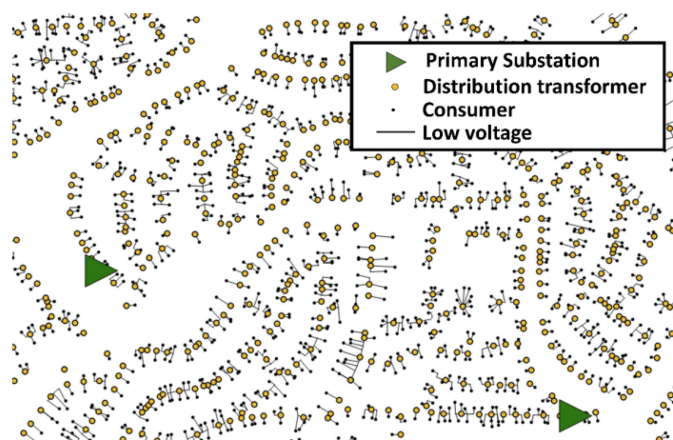


Fig. 12. Primary substations.

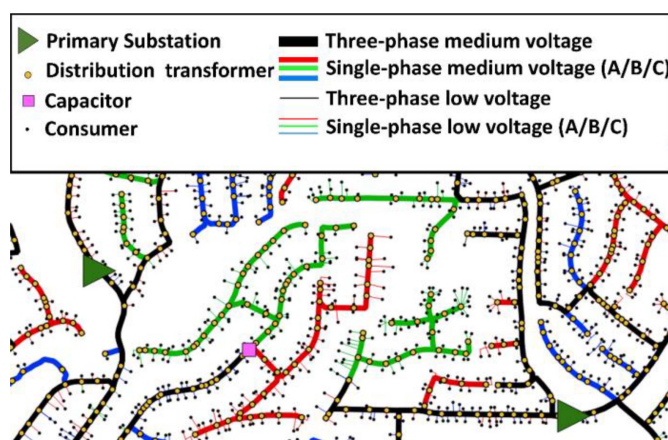


Fig. 14. Single-phase, three-phase feeders, and capacitors.

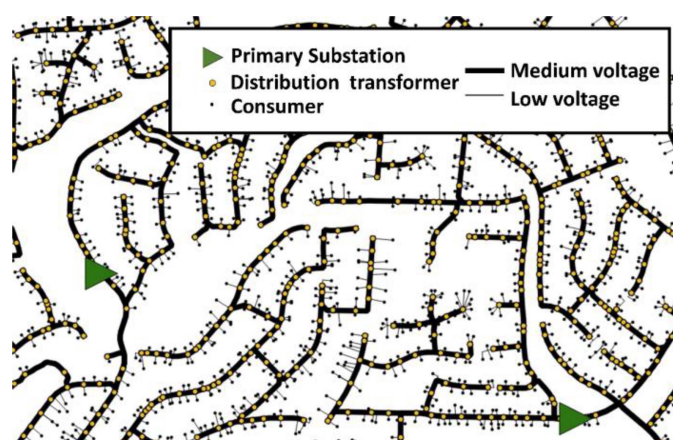


Fig. 13. Medium voltage system.

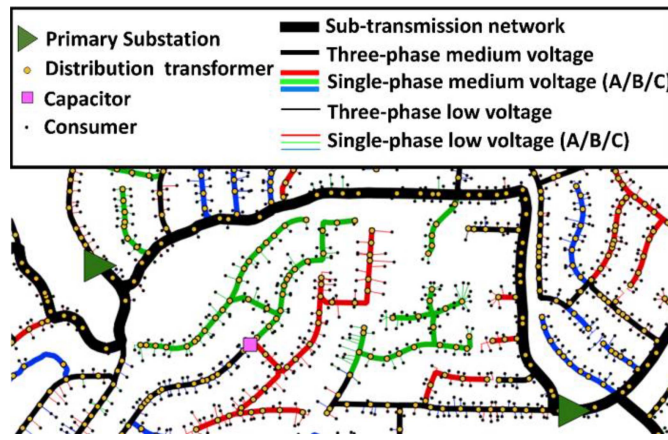


Fig. 15. Sub-transmission system.

Once the primary substations and the distribution transformers are placed and sized, the MV system is generated, as shown in Fig. 13. After applying a minimum spanning tree to obtain the graph that connects all the nodes [16], branch-exchange algorithms are applied to solve technical constraints [17] and meet target metrics. Every time a power line that connects two nodes is planned, the algorithms interact with the GIS layer of potential service drops to obtain a realistic layout connecting the two nodes and comply with the street map.

Initially, all the conductors have the maximum allowed ampacity and are three-phase. The next step is deciding single-phase laterals. The model makes this decision by minimizing the cost of the system while considering the costs of the conductors and energy losses. In this stage, each single-phase line section is assigned to phases A, B, and C, with the objective of minimizing the unbalance, as explained in [27]. Fig. 14 shows phases A, B, and C, representing them by red, green, and blue, respectively. Capacitors and voltage regulators are also placed to improve voltages.

Finally, the sub-transmission system is designed. It connects the primary substations to the transmission substations. In Fig. 15, the thicker black lines correspond to the sub-transmission system, connecting the two primary substations that supply this particular distribution area.

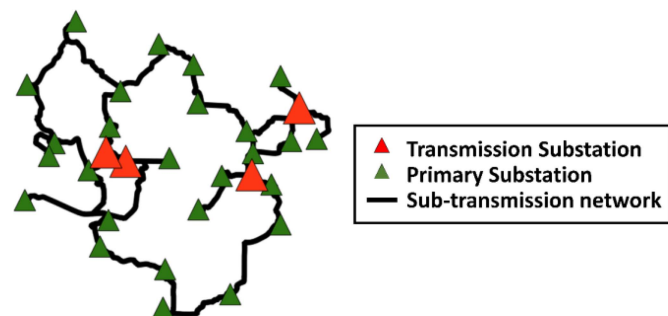


Fig. 16. Sub-transmission system in the urban area (zoom-out).

Fig. 16 shows the sub-transmission system that supplies the whole urban-suburban area being planned, whereas Fig. 15 shows only a zoom-in.

Table I shows the summary statistics of the full distribution system built for the GSO synthetic system. It covers three distribution areas: urban-suburban, industrial, and rural, supplying 92,550 consumers, with 23 primary substations, 17,803 distribution transformers, and 4,584 km of system length. This distribution system was built with RNM-US using an Intel Core i7-6700 CPU 3.4 GHZ, running separately the urban, industrial, and rural areas to allow setting different targets for each of them. The urban area is significantly larger,

TABLE I
SUMMARY STATISTICS FOR GSO SYNTHETIC SYSTEM (v0.2)

Metric	Urban-Suburb.	Industrial	Rural	Total
Number of buildings	60,924	2,834	6,350	70,108
Medium voltage	43	36	0	79
Low voltage	60,880	2,797	6,350	70,027
Number of consumers	82,176	4,024	6,350	92,550
Medium voltage	335	36	0	371
Low voltage	81,841	3,988	6,350	92,179
Number of buses	126,554	6,644	14,984	148,182
Number of electrical nodes	277,777	16,789	32,276	326,842
Number of transmission substations	2	1	1	4
Number of primary substations	17	4	2	23
Transformers	13,662	1,377	2,764	17,803
Length of power lines (km)	3,528	303	754	4,584
Sub-transmission	69	15	12	96
Medium voltage	1,194	192	410	1,796
Low voltage	2,265	95	332	2,692
Number of primary feeders	100	21	9	130

requiring 145 min of computing time, whereas the industrial and rural areas require only 4 min and 7 min, respectively. This system is available to the research community from the DR POWER [42] and Bettergrids.org [43] repositories in OpenDSS, CYME, and GIS formats. The references include the specific URLs within the repositories, where the synthetic distribution networks are available. The input consumers and street map are available in the repositories in GIS format. A summary of the catalog of equipment is included in the Appendix, and the full catalog is available in [44].

B. Very Large-Scale Synthetic System: San Francisco Bay Area

The RNM-US is also used to obtain a very large-scale system. In particular, 12 Californian counties are modeled: Lake, Sonoma, Napa, Solano, Marin, San Francisco, Contra Costa, Alameda, San Mateo, Santa Cruz, Santa Clara, and San Benito. The area extends about 26,000 km² (10,000 mi²).

The full synthetic SFO system is illustrated in Fig. 17. The dataset was built with RNM-US using an Intel Xeon CPU E5-2660 v3 2.6 GHz, with 20 cores and 40 logical processors. 106 CPU hours are required; however, the elapsed time is only 12.5 hours by using several processors in parallel. This system is also available from the DR POWER [42] and Bettergrids.org [43] repositories. The characteristics of the system are diversified in different regions in terms of different voltage classes (4 kV, 12.47 kV, and 25kV), several types of voltage management (regulators vs. capacitors), two primary substation arrangements (delta-wye and delta-delta), and different types of areas.

The summary statistics of the SFO system are shown in Table II. The SFO systems covers 4.3 million consumers, 9.8 million electrical nodes, 632 primary substations, 559,151 distribution transformers, and 116,837 km of power lines. Despite the very large size of this system, the distribution

TABLE II
SUMMARY STATISTICS FOR SFO SYNTHETIC SYSTEM (v0.8)

Metric	Value
Number of buildings	2,265,594
Medium voltage	1,535
Low voltage	2,264,014
Number of consumers	4,299,805
Medium voltage	11,503
Low voltage	4,288,297
Number of buses	4,916,869
Number of electrical nodes	9,868,205
Number of transmission substations	148
Number of primary substations	632
Number of distribution transformers	559,151
Length of power lines (km)	116,837
Sub-transmission	4,128
Medium voltage	64,460
Low voltage	48,249
Number of feeders	2,236

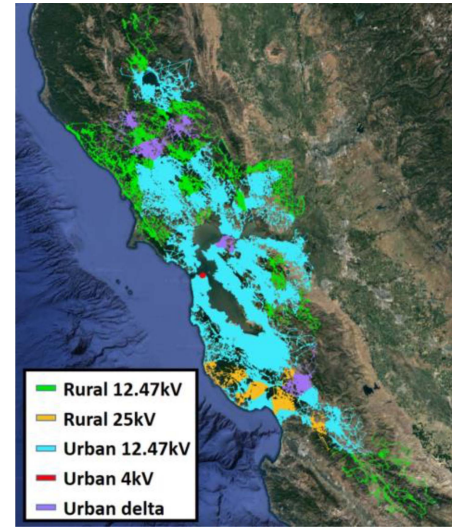


Fig. 17. Synthetic distribution system in San Francisco Bay Area.

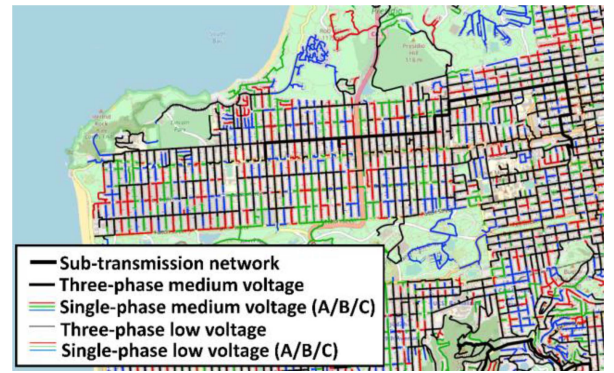


Fig. 18. Zoom-in of the synthetic distribution system in San Francisco.

system is very detailed, as shown in Fig. 18, where a zoom-in of the synthetic power system in San Francisco downtown is superimposed on the street map.

C. Validation of Synthetic Systems

Details on the validation process and results for the synthetic distribution systems presented in Sections VI-A and VI-B are documented in [19].

TABLE III
CATALOG OF POWER LINES

Voltage level	Voltage (kV)	Number of phases	R (Ohms/km)	X (Ohms/km)	C (nF/km)	R ₀ (Ohms/km)	X ₀ (Ohms/km)	C ₀ (nF/km)
LV	0.12 - 0.24	1	0.26 - 1.4	0.09 - 0.1	53 - 62	0.85 - 2.55	0.46 - 1.81	2 - 10
LV	0.48	3	0.08 - 0.3	0.22 - 0.24	179 - 311	0.53 - 0.79	1.1 - 1.2	179 - 311
MV	4 - 12.47 - 25	1	0.18 - 1.0	0.54 - 0.95	8 - 217	0.18 - 1.04	0.54 - 0.95	8 - 217
MV	4 - 12.47 - 25	2	0.12 - 0.3	0.42 - 0.61	10	0.25 - 0.53	0.18 - 1.8	3 - 5
MV	4 - 12.47 - 25	3	0.06 - 0.6	0.21 - 0.61	10 - 339	0.15 - 0.54	0.21 - 2.24	4 - 339
HV	69	3	0.12 - 0.6	0.24 - 0.4	11 - 279	0.24 - 0.37	0.97 - 2.07	6 - 279

TABLE IV
CATALOG OF DISTRIBUTION TRANSFORMERS

Primary voltage (kV)	Number of phases	Installed power (kVA)	Reactance (p.u.)	No load losses (kW)	Load losses (kW)
4 - 12.47 - 25	1	10 - 75	0.009 - 0.02	0.07 - 0.29	0.08 - 0.3
4 - 12.47 - 25	3	75 - 1500	0.032 - 0.057	0.45 - 2.88	1.31 - 21

VII. CONCLUSION

This paper provides an overview of the algorithms used to generate synthetic distribution systems using RNM-US. The system components include power lines, substations, distribution transformers, voltage regulators, capacitors, and switching devices. Heuristics methods are used to plan very large-scale distribution areas, covering millions of consumers and hundreds of thousands of distribution transformers. At the same time, the systems are described in detail, modeling equipment of all the voltage levels and allowing the use of power system analysis tools. The use of street maps enables obtaining realistic results resembling actual distribution systems. The validation process (described in [19]) is used to calibrate the RNM-US algorithms. The paper presents two large-scale systems: GSO and SFO, both of which have been released freely for public use.

A future line of research is coupling very large-scale distribution systems with transmission systems to obtain combined transmission and distribution system models.

APPENDIX

The catalog of equipment includes 209 power lines, 52 distribution transformers, 21 primary substations, 3 transmission substations, 6 types of switching devices, and 2 voltage management equipment. Tables III and IV summarize the main parameters of the power lines and distribution transformers, respectively. The substation configurations are more complex and are added in the post-processing stage. The full catalog is available in [44].

ACKNOWLEDGMENT

The views expressed in the article do not necessarily represent the views of the U.S. Government. The publisher, by accepting the article for publication, acknowledges that the U.S. Government retains a nonexclusive, paid-up, irrevocable, worldwide license to publish or reproduce the published form of this work, or allow others to do so, for U.S. Government purposes.

REFERENCES

- [1] F. E. P. Marcos *et al.*, "A review of power distribution test feeders in the United States and the need for synthetic representative networks," *Energies*, vol. 10, no. 11, p. 1896, 2017.
- [2] R. F. Arritt and R. C. Dugan, "The IEEE 8500-node test feeder," in *Proc. IEEE PES Transm. Distrib. Conf. Expo.*, 2010, p. 1.
- [3] C. Mateo, T. Gómez, Á. Sánchez-Miralles, J. P. Peco, and A. Candela, "A reference network model for large-scale distribution planning with automatic street map generation," *IEEE Trans. Power Syst.*, vol. 26, no. 1, pp. 190–197, Feb. 2011.
- [4] T. Gómez, C. Mateo, Á. Sánchez, P. Frías, and R. Cossent, "Reference network models: A computational tool for planning and designing large-scale smart electricity distribution grids," in *High-Performance Computing in Power and Energy Systems*, S. K. Khaitan, A. Gupta, S. Aluru, and K. Gopalakrishnan, Eds. Heidelberg, Germany: Springer Verlag, 2012.
- [5] L. Pieltain, T. Gómez, R. Cossent, C. Mateo, and P. Frías, "Assessment of the impact of plug-in electric vehicles on distribution networks," *IEEE Trans. Power Syst.*, vol. 26, no. 1, pp. 206–213, Feb. 2011.
- [6] R. Cossent, T. Gómez, L. Olmos, C. Mateo, and P. Frías, "Assessing the impact of distributed generation on distribution network costs," in *Proc. 6th Int. Conf. Eur. Energy Market*, 2009, pp. 586–593.
- [7] M. Vallés, J. Reneses, P. Frías, and C. Mateo, "Economic benefits of integrating active demand in distribution network planning: A Spanish case study," *Elect. Power Syst. Res.*, vol. 136, p. 331, Jul. 2016.
- [8] H. M. Khodr, Z. Vale, C. Ramos, and P. Faria, "Optimization techniques for power distribution planning with uncertainties: A comparative study," in *Proc. IEEE Power Energy Soc. Gen. Meeting*, 2009, pp. 1–8.
- [9] R. Li, W. Wang, Z. Chen, J. Jiang, and W. Zhang, "A review of optimal planning active distribution system: Models, methods, and future researches," *Energies*, vol. 10, no. 11, p. 1715, 2017.
- [10] P. S. Georgilakis and N. D. Hatziaargyriou, "A review of power distribution planning in the modern power systems era: Models, methods and future research," *Elect. Power Syst. Res.*, vol. 121, pp. 89–100, Apr. 2015.
- [11] M. Resener, S. Haffner, L. A. Pereira, and P. M. Pardalos, "Optimization techniques applied to planning of electric power distribution systems: A bibliographic survey," *Energy Syst.*, vol. 9, no. 3, pp. 473–509, 2018.
- [12] S. M. Mazhari and H. Monsef, "A hybrid heuristic and evolutionary algorithm for distribution substation planning," *IEEE Syst. J.*, vol. 9, no. 4, pp. 1396–1408, Dec. 2015.
- [13] P. Wen and L. Wenxia, "Niche genetic algorithm and minimum spanning tree for substation planning," in *Proc. WRI Global Congr. Intell. Syst.*, vol. 2, 2009, pp. 61–65.
- [14] A. Navarro and H. Rudnick, "Large-scale distribution planning—Part I: Simultaneous network and transformer optimization," *IEEE Trans. Power Syst.*, vol. 24, no. 2, pp. 744–751, May 2009.
- [15] L. González-sotres, C. Mateo, Á. Sánchez-miralles, and M. A. Miró, "Large-scale MV/LV transformer substation planning considering network costs and flexible area decomposition," *IEEE Trans. Power Del.*, vol. 28, no. 4, pp. 2245–2253, Oct. 2013.
- [16] S. Najafi Ravadanegh, A. Vahidnia, and H. Hatami, "On optimal design and expansion of electrical power distribution systems," *J. Circuits, Syst. Comput.*, vol. 19, no. 1, pp. 45–58, 2010.

- [17] J. Salehi and M. R. Haghifam, "Long term distribution network planning considering urbanity uncertainties," *Int. J. Elect. Power Energy Syst.*, vol. 42, no. 1, pp. 321–333, 2012.
- [18] M. Sedghi, M. Aliakbar-Golkar, and M. R. Haghifam, "Distribution network expansion considering distributed generation and storage units using modified PSO algorithm," *Int. J. Elect. Power Energy Syst.*, vol. 52, pp. 221–230, Nov. 2013.
- [19] V. Krishnan *et al.*, "Validation of synthetic U.S. power distribution system datasets," *IEEE Trans. Smart Grid*, early access, Mar. 20, 2020. [Online]. Available: <https://doi.org/10.1109/TSG.2020.2981077>, doi: 10.1109/TSG.2020.2981077
- [20] G. M. dos Santos Pereira, T. S. P. Fernandes, and A. R. Aoki, "Allocation of capacitors and voltage regulators in three-phase distribution networks," *J. Control Autom. Elect. Syst.*, vol. 29, no. 2, pp. 238–249, 2018.
- [21] A. Rodriguez-Calvo, P. Frias, J. Reneses, and C. Mateo, "Optimal degree of smart transformer substations in distribution networks for reliability improvement," in *Proc. 3rd IEEE PES Int. Conf. Exhibit. Innov. Smart Grid Technol. (ISGT Europe)*, 2012, p. 1.
- [22] E. Schweitzer, A. Scaglione, A. Monti, and G. A. Pagani, "Automated generation algorithm for synthetic medium voltage radial distribution systems," *IEEE J. Emerg. Sel. Top. Circuits Syst.*, vol. 7, no. 2, pp. 271–284, Jun. 2017.
- [23] J. C. Moreira, E. Míguez, C. Vilachá, and A. F. Otero, "Large-scale network layout optimization for radial distribution networks by parallel computing: Implementation and numerical results," *IEEE Trans. Power Del.*, vol. 27, no. 3, pp. 1468–1476, Jul. 2012.
- [24] G. Pisano *et al.*, "Synthetic models of distribution networks based on open data and georeferenced information," *Energies*, vol. 12, no. 23, p. 4500, 2019.
- [25] C. K. Gan, P. Mancarella, D. Pudjianto, and G. Strbac, "Statistical appraisal of economic design strategies of LV distribution networks," *Elect. Power Syst. Res.*, vol. 81, no. 7, pp. 1363–1372, 2011.
- [26] S. S. Saha, E. Schweitzer, A. Scaglione, and N. G. Johnson, "A framework for generating synthetic distribution feeders using openstreetmap," in *Proc. 51st North Amer. Power Symp. (NAPS)*, 2019, p. 6.
- [27] F. Postigo *et al.*, "Phase selection algorithms to minimize cost and imbalance in U.S. synthetic distribution networks," *Int. J. Elect. Power Energy Syst.*, vol. 120, Sep. 2020, Art. no. 106042.
- [28] M. Grzanic, M. G. Flammini, and G. Pretico, "Distribution network model platform: A first case study," *Energies*, vol. 12, no. 21, p. 4079, 2019.
- [29] M. B. Larsson, "The network performance assessment model," Licent. thesis, Dept. Elect. Eng., Roy. Inst. Technol., Stockholm, Sweden, Feb. 2005.
- [30] J. A. G. Conejo, G. T. Bartolome, A. G. Exposito, and M. R. Montanes, "ANETO: A system for the automatic generation of theoretical network models," in *Proc. 9th Int. Conf. Elect. Power Qual. Util.*, Oct. 2007, pp. 1–504.
- [31] J. Carr and L. V. McCall, "Divergent evolution and resulting characteristics among the world's distribution systems," *IEEE Trans. Power Del.*, vol. 7, no. 3, pp. 1601–1609, Jul. 1992.
- [32] C. Mateo *et al.*, "European representative electricity distribution networks," *Int. J. Elect. Power Energy Syst.*, vol. 99, pp. 273–280, Jan. 2018.
- [33] R. Hendron and C. Engebrecht, *Building America House Simulation Protocols*. Washington, DC, USA: U.S. Dept. Energy, 2010, p. 79.
- [34] M. Deru *et al.*, *U.S. Department of Energy Commercial Reference Building Models of the National Building Stock*, U.S. Dept. Energy, Washington, DC, USA, Feb. 2011, pp. 1–118.
- [35] R. C. Dugan and T. E. McDermott, "An open source platform for collaborating on smart grid research," in *Proc. IEEE Power Energy Soc. Gen. Meeting*, 2011, p. 7.
- [36] L. Bam and W. Jewell, "Review: Power system analysis software tools," in *Proc. IEEE Power Eng. Soc. Gen. Meeting*, 2005, pp. 146–151.
- [37] J. D. McDonald, B. Wojczech, B. Flynn, and I. Voloh, *Encyclopedia of Sustainability Science and Technology*. New York, NY, USA: Springer, 2012.
- [38] M. Barthélemy, "Spatial networks," *Phys. Rep.*, vol. 499, nos. 1–3, pp. 1–101, 2011.
- [39] D. F. Pires, A. G. Martins, and C. H. Antunes, "A multiobjective model for VAR planning in radial distribution networks based on tabu search," *IEEE Trans. Power Syst.*, vol. 20, no. 2, pp. 1089–1094, May 2005.
- [40] T. H. Chen, W. T. Huang, J. C. Gu, G. C. Pu, Y. F. Hsu, and T. Y. Guo, "Feasibility study of upgrading primary feeders from radial and open-loop to normally closed-loop arrangement," *IEEE Trans. Power Syst.*, vol. 19, no. 3, pp. 1308–1316, Aug. 2004.
- [41] H. Edelsbrunner and N. R. Shah, "Incremental topological flipping works for regular triangulations," *Algorithmica*, vol. 15, no. 3, pp. 223–241, 1996.
- [42] *DR Power*. Accessed: Jan. 5, 2020. [Online]. Available: https://eGRIDdata.org/group/national-renewable-energy-laboratory?sort_by=changed&f%5B0%5D=field_tags%3A693
- [43] *BetterGrids.org: A Standards-Based Intelligent Repository for Collaborative Grid Model Management*. Accessed: Jan. 5, 2020. [Online]. Available: <https://db.bettergrids.org/bettergrids/handle/1001/94>
- [44] C. Mateo *et al.* (2020). *RNM-US Catalog of Equipment, IEEE Dataport*. Accessed: Jun. 18, 2020. [Online]. Available: <http://dx.doi.org/10.21227/4vne-pd37>



Carlos Mateo received the Ph.D. degree in industrial and computer engineering from Comillas Pontifical University, Madrid, Spain, in 2007, where he is a Member of the Institute for Research in Technology, School of Engineering (ICAI). His current research interests include modeling, simulation, and algorithms, especially in the fields of electricity distribution networks and distributed energy resources.



Fernando Postigo received the M.S. degree in industrial engineering with a specialization in electricity from Comillas Pontifical University, Madrid, Spain, in 2016, where he is currently pursuing the Ph.D. degree in power systems with the Institute for Research in Technology. His research focuses on the development of tools for the modeling and analysis of distribution networks in the context of increasing penetrations of renewables.



Fernando de Cuadra received the B.S. degree, the M.S. degree, and the Ph.D. degree in industrial engineering from the ICAI School of Engineering, Universidad Pontificia Comillas, Madrid, Spain, and an Executive M.B.A. degree from the ICADE Business School, Universidad Pontificia Comillas, where he has been a Full Professor with the ICAI and IIT since 2001. In 2016, he joined the Universal Energy Access Lab, a joint research group of MIT and IIT. Among other positions, he was appointed as the Dean of the ICAI School of Engineering from 2001 to 2010. His main research areas are related to Operations Research techniques (large-scale simulation and direct-search optimization) applied to sectors such as aerospace, railways, and power systems.



Tomás Gómez San Roman (Senior Member, IEEE) received the Ph.D. degree in industrial engineering from the Polytechnic University of Madrid, Spain, in 1989. He is a Professor of electrical engineering with the Engineering School, Comillas Pontifical University. From 2011 to 2013, he served as a Commissioner with the Spanish Energy Regulatory Agency. He has broad industrial experience in joint research projects in the field of electric power systems. His areas of interest are the operation and planning of transmission and distribution systems, power quality assessment and regulation, and economic and regulatory issues in the electric power sector.



Tarek Elgindy received the B.S. degree from the University of Sydney and the M.S. degree in algorithms, combinatorics, and optimization from Carnegie Mellon University. He is an Engineer with the Grid-Connected Energy systems Modeling Group, National Renewable Energy Laboratory. He has worked extensively on managing large electrical distribution data sets and understanding the impact of network designs on power quality primarily for quasi-static time-series analysis. His research interests include developing market structures for

distribution networks, developing synthetic electrical infrastructure data, and understanding the interactions between distribution and transmission systems with high penetrations of distributed energy resources.



Venkat Krishnan (Senior Member, IEEE) received the M.S. and Ph.D. degrees in electrical engineering from Iowa State University in 2007 and 2010, respectively. He is a Senior Engineer with the Sensing and Predictive Analytics Group, Power Systems Engineering Center, National Renewable Energy Laboratory. His expertise is in power system operations (markets and variable renewable integration), transmission and distribution grid stability assessment, energy storage integration, long-term capacity expansion planning, and the application of

statistical simulations and data analytics methods in power systems.

Pablo Dueñas received the B.Sc., M.Sc., and Ph.D. degrees in electrical engineering from Comillas Pontifical University, where he is a Research Scientist with MIT Energy Initiative. In 2014, he joined the Massachusetts Institute of Technology. His research focuses on the technical, economic, and regulatory aspects of interdependent natural gas and electricity systems, energy security under carbon-constrained energy policies, and the future of distributed energy resources.



Bri-Mathias Hodge (Senior Member, IEEE) is an Associate Professor with the Department of Electrical, Computer and Energy Engineering and a Fellow of the Renewable and Sustainable Energy Institute, University of Colorado Boulder. He is also Chief Scientist with the Power Systems Engineering Center, National Renewable Energy Laboratory.



Bryan Palmintier (Senior Member, IEEE) the undergraduation degree in mechanical engineering from Stanford University, the B.S. degree in aerospace engineering from the Georgia Institute of Technology, the M.S. degree in aero/astro engineering from Stanford University, and the Ph.D. degree in engineering systems from the Massachusetts Institute of Technology. He is a Manager of the Grid-Connected Energy systems Modeling Group and a Principal Research Engineer with the National Renewable Energy Laboratory.

His current research focuses on advanced integrated analysis methods for renewable and distributed energy systems.